



Housing Insecurity Among Transgender Individuals: Racially Heterogeneous Gender Effects in the Household Pulse Survey

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Abstract

This study aims to compare differences in housing cost burden between transgender and cisgender populations in the USA using data from the Household Pulse Survey. To assess the housing cost burden, I construct a variable that reflects the range within which the cost burden—defined as the ratio of rent to income—falls for each observation. I conceptualize “gendered rent premia” as the increased rent burden faced by transgender individuals relative to cisgender men. On average, households headed by transgender men experience a 7.9% higher rent burden, while households led by individuals who do not identify as either women or men face a 9.5% higher rent burden. These gendered rent premia are observed across both residents and non-residents of the 15 most populous metropolitan statistical areas and within households consisting of one or two adults. Additionally, transgender women’s households also experience rent premia in households with either one or two adults. The analysis further reveals that gendered rent premia exhibit racial variation, highlighting significant disparities in rent burdens across different racial groups within the transgender population.

Keywords Transgender · Racial heterogeneity · Rental housing · Discrimination

Introduction

Although there has been an expanding body of literature within the field of economics focused on the economic well-being of transgender individuals, there remains a notable gap in the research concerning their interactions, in the USA, with the housing market. Existing studies suggest that transgender individuals face significant social and economic barriers (Badgett et al. 2024, 2021), which may hinder their ability to acquire and maintain stable housing.¹ Many transgender people report discrimination has worsened their housing security. Of respondents to the 2015 U.S. Transgender Survey, 23% reported experiences of housing discrimination within the previous year, and 12% indicated they had experienced homelessness during the same period due to their transgender status (James et al. 2016).

Within the broader economics literature, the establishment of evidence for discrimination continues to be a contentious

issue. However, this study does not aim to provide direct evidence of discrimination, as is typically done in the economics of discrimination literature, as it focuses on rental housing allocation outcomes rather than the processes that drive those outcomes. Nevertheless, the findings of this study suggest that transgender individuals face economic challenges to securing housing that are consistent with discrimination.

This paper represents, to my knowledge, the first quantitative analysis of the relationship between transgender individuals and the rental housing market in the USA. It employs data from the nationally representative Household Pulse Survey to demonstrate that, relative to their cisgender counterparts, transgender individuals and their households are required to allocate a larger proportion of their income to meet rental housing costs.

Literature Review

The Economic Status of Transgender People

While long understudied, in recent years a vibrant literature on the economic status of transgender people has flourished within Economics. The emerging literature on the economic

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¹ This paper will discuss transgender individuals, who have genders which are not the gender they were assigned at birth. Cisgender individuals have genders which are those they were assigned at birth.

status of transgender people has developed consensus findings: transgender people earn lower incomes than similar cisgender people (Carone et al. 2021; Carpenter et al. 2020, 2022; Shannon 2022).

Most economists frame discrimination within markets through a narrow lens, with much of the debate centered around the economic rationality of such behavior. Becker theorized discriminatory behavior within markets as arising from utility-maximizing behavior from actors with a “taste for discrimination” and argued that such behavior will be punished in competitive markets by non-discriminatory actors (Becker 2010).

In addition to discriminatory preferences, information asymmetry is another way economists theorize the emergence of discrimination. Statistical discrimination occurs when economic agents rely on heuristics or other shortcuts, such as race or transgender identity, as proxies for more detailed information.

A growing body of economic literature has emerged which finds evidence of labor market discrimination directed toward transgender people. Granberg et al. (2020) find that transgender job applicants in Sweden receive fewer positive responses to their applications. Similarly, Drydakis (2024) finds evidence of discrimination against transgender women seeking employment after college, with a positive reception most common among low-wage positions. The literature is not uniform in finding evidence of labor market discrimination. Though Kline et al. (2022) find evidence of systemic racial and gender bias among large US firms, they do not find statistically significant evidence that gender-neutral pronouns, an indicator of nonbinary (and, to an extent, transgender) gender identity, are associated with hiring discrimination. The authors state, however, that their study design limits the statistical precision of this estimation; fewer than ten percent of resumes contained pronoun information (Kline et al. 2022).

Why might transgender people face discrimination in labor markets? Campbell et al. (2024) finds that transgender women with voices perceived to be more masculine were less likely to be employed. Discontinuity between individual gender expression and legal documents signifying gender may also contribute. Mann (2021) finds a negative relationship between the employment rates of transgender women and the difficulty associated with changing their gender markers on legal documents. As an emerging field, the study of labor market discrimination against transgender people is, for now, limited to a handful of published articles.

Economic research on the relationship between transgender people and the housing markets is sparse, but the existing literature suggests that transgender individuals face pervasive housing insecurity, defined by frequent changes in housing and homelessness. Qualitative research, such as that produced by Glick et al. (2019, 2020), indicates that transgender people face great difficulty acquiring and retaining stable

housing. Of the transgender respondents to the Transpop and Generations surveys, 20% reported moving more than three times in the previous two years (Meyer et al. 2021). During the COVID-19 pandemic, transgender individuals reported experiencing housing insecurity or homelessness at more than twice the rate of cisgender individuals (Felt et al. 2023).

Discrimination in Housing

Experimental evidence suggests that racial and ethnic discrimination in housing provision is widespread and harms members of marginalized populations. Ondrich et al. (1999), using an early audit experiment, found evidence of discrimination against Black or Latine applicants in rental housing markets. Later, Hanson and Hawley (2011), using a correspondence experiment, confirm this earlier finding, and they show evidence that discrimination is lessened against Black applicants who indicate high social class. Discrimination worsens against applicants in large apartment buildings and in neighborhoods near “tipping points” where racially dominant people make up between 80 and 95% of the population. In a meta-analysis of rental housing correspondence tests performed between 2006 and 2017 in OECD countries, Flage (2018) finds consistent evidence of discrimination against ethnic minorities, with private landlords engaging in statistical discrimination more often than real estate agents.

In a similar vein, gender discrimination in housing markets remains understudied but discriminatory behavior is found to be heterogeneous across racial or ethnic categories. Flage (2018) finds evidence that ethnic discrimination within the rental housing market is gendered. Women experience less discrimination than men. Ethnic minority women see a reduction in discrimination compared to ethnic minority men which is larger than the difference between ethnic majority men and women. When applying for mortgages, lenders reinforce dominant racial-gender norms by discriminating against white couples with working women and Black or Latine couples where women did not work (Robinson 2002).

Research on housing discrimination against LGBTQ+ people, though a small field, has found consistent results. In a correspondence study of the Swedish rental housing market, a same-sex couple received fewer responses than a heterosexual one, indicating discrimination against same-sex couples (Ahmed and Hammarstedt 2009). Similar results are found with correspondence studies in the USA (Friedman et al. 2013; Schwegman 2019). All three studies provide only limited information on the scale of discrimination against LGBTQ+ individuals by studying discrimination that affects same-sex couples; such findings are not directly applicable to transgender or non-binary / gender non-conforming individuals.

There is an emerging literature which uses correspondence methods to identify and diagnose discrimination in the hous-

ing market against transgender people. Statistical analysis using an audit study found evidence that transgender individuals, otherwise identical to cisgender individuals, were less likely to have the opportunity to rent housing (Levy et al. 2017). Jansson and Fritzson (2022) found, in the Swedish rental housing market, that applications which implied the applicant was transgender received fewer invitations for further contact. In four Latin American countries, Argentina, Colombia, Ecuador, and Peru, researchers using a correspondence study found evidence of statistical discrimination against couples that contain a cisgender man and a transgender woman (Abbate et al. 2024).

Discrimination in Housing Markets: A Conceptual Framework

Due to the intrinsic connection between housing provision and income acquisition, discriminatory practices in either area may influence the market allocation of housing to marginalized populations. Recognition of the interconnection between discriminatory practices in one sector and disparate outcomes in another is a cornerstone of the recent working paper by Bohren et al. (2022), which decomposes discrimination in the labor market into both direct and indirect effects. Although Bohren et al. (2022) offers greater clarity compared to many economic analyses of discrimination, I offer a more concrete model which identifies two primary pathways through which discrimination may alter housing provision: discrimination in labor markets or in housing markets may influence housing provision via their effects on housing affordability. This analysis will focus on the provision of rental housing, due to data availability. Housing affordability is measured using the rent burden of the household. Rent burden is the ratio of household rental payments divided by household income:

$$\text{rent burden} = \frac{\text{rental payments}}{\text{income}} \quad (1)$$

In housing markets, individuals or households may directly face discrimination due to their identity, leading to steering, housing application denial, or hedonic pricing. Steering is the process by which minoritized individuals may be primarily shown housing in neighborhoods with individuals with similar marginalized identities by real estate agents and landlords (Christensen and Timmins 2022).² All three of these practices can lead to increased rental housing payments for households.

² Minorities, here, refers to individuals who may wield less political, social, and economic power, compared to individuals and groups aligned with dominant social identities.

To understand this, consider the following bargaining model, adapted from the discussion of price discrimination in the housing market offered by Ihlanfeldt and Mayock (2009). Although competitive markets may lead to the elimination of discrimination or prejudice as a persistent cause of differences in market outcomes (Becker 2010), a bargaining model more closely approximates the reality of markets for housing or labor: they are characterized by heterogeneous bargaining power and heterogeneous goods. As such, prejudice by individuals with greater bargaining power would not be eliminated and would harm the subjects of prejudice (Ihlanfeldt and Mayock 2009).

Imagine the process by which a landlord and potential renter bargain over the rental price of an apartment. Both the landlord and potential tenant must agree upon the price and either may refuse an offer made by the other. The decisions of both parties, and thus the rental payment for the unit, will follow from their individual utility functions and from their possible fallback positions, should they fail to come to an agreement. For this bargaining process, then, variation in fallback position as well as the structure of individual utility functions may result in differences in rental payments.

Both steering and increased rates of housing application denial function to weaken the relative bargaining strength of minorities. Households steered to specific neighborhoods find that the effective size of the housing market they may access is limited. Though this may either strengthen the position of the landlord or weaken that of a potential tenant, the result will be the same: steering can put upward pressure on rental housing prices. Potential tenants who are more likely to find their application denied will face a worse relative bargaining position as well and will be more likely to accept higher rental payments to secure housing.³

Although steering and differences in rates of application denial may influence bargaining between landlords and potential tenants by changing relative fallback positions, discrimination may also influence housing market bargaining through the utility functions of participants. A landlord may incorporate discriminatory tastes and preferences into their bargaining behavior. These tastes may be their own, or they may reflect the biases of other customers if the landlord expects an agreement with a potential tenant to impact their other business relationships. These discriminatory prefer-

³ Evidence, such as that provided by Rosen et al. (2021), points towards differences in scale as an important determinant of the mechanism used to weaken the bargaining position of minorities in housing markets. For example, smaller-scale landlords rely upon stereotyping to screen out minority applicants, in a textbook example of statistical discrimination. Larger landlords, relying upon algorithms which do not directly consider applicant minority status, may also disproportionately deny applications submitted by minorities if those minorities tend to have lower incomes or less favorable credit histories. These lower incomes or worse credit histories may, themselves, reflect discrimination.

ences, when incorporated into the preferences of the landlord, manifest as an increased price, where the increased utility arising from the greater rental revenue compensates for the disutility which arises from engaging with the tenant.

Whether through manipulating fallback positions or hedonic pricing, discriminatory practices in housing markets can manifest as price differentials conditional upon minority status. Empirical evidence supports this. Ihlanfeldt and Mayock (2009) find evidence that discrimination may appear as mark-ups (mark-downs) offered to minorities (majorities) in home sales.⁴ This finding is found again in the home ownership market (Bayer et al. 2017; Cattaneo and Feir 2021) and in the rental market (Early et al. 2019). Although this literature finds price discrimination in the context of racial discrimination, the implicitly identified mechanism can easily operate in other bargaining frameworks with power asymmetry and heterogeneous goods.

Figure 1 illustrates the pathways through which discrimination can lead to increased rent burdens among transgender people. These pathways can occur within either labor or rental housing markets. Statistical discrimination may also lead to “self-fulfilling prophecy” effects. A self-fulfilling prophecy effect is a positive-feedback pathway through which incorrect prior beliefs, resulting from information asymmetry, are made to appear true due to the behavior of participants (Merton 1948). The two pathways and the influence of self-fulfilling prophecy effects on rent burden will be explained in further detail below Fig. 1.

The first of these pathways occurs through the labor market. Discrimination against transgender people may limit their housing affordability through the labor market by lowering their earnings, which then increases the relative cost of housing as a proportion of their income.

The second of these pathways occurs through the housing market. Within the housing market, discrimination by sellers of housing or by landlords may increase the relative cost of housing. Landlords or house sellers with Beckerian “tastes for discrimination” may demand compensation in the form of higher rent payments or higher home prices (and thus a larger mortgage payment) when engaging with transgender individuals. This would increase the relative cost of housing for those transgender individuals.

The self-fulfilling prophecy effects occur, like these two pathways, in both the labor and rental housing markets, but an examination of the self-fulfilling prophecy dynamics of the labor market is beyond the scope of this paper. Within the rental housing market, landlords may also face imperfect

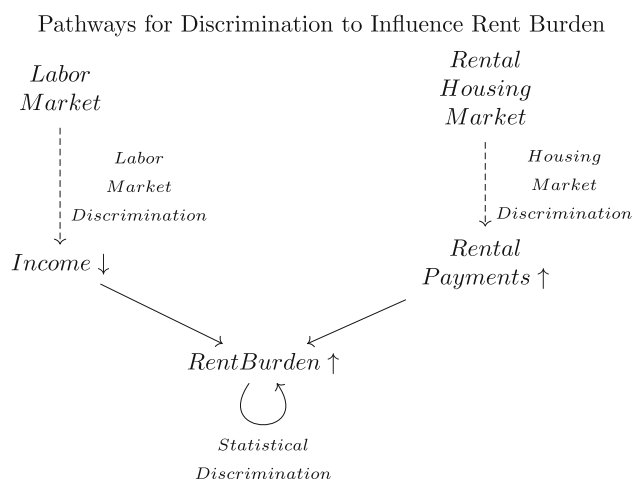


Fig. 1 Pathways for discrimination to influence rent burden. The diagram above illustrates the two routes through which discrimination against transgender people could manifest as an increased rent burden. Solid black lines indicate causal direction. Dashed black lines show which mechanisms are tied to particular pathways through which discrimination can influence rent burden. An up arrow, ↑, indicates a value is rising while a down arrow, ↓, indicates the opposite

information, and thus engage in statistical discrimination, refusing to rent to transgender individuals. This statistical discrimination may result from incorrect assessments of a transgender individual’s ability to pay for housing. If so, such discrimination would leave transgender individuals with decreased bargaining power and thus they would be more willing to purchase or rent housing at a higher price than desired. Such a situation, with systematically weakened bargaining positions for transgender individuals, would increase the relative cost of housing for transgender individuals, as landlords are able to bargain from a position of greater relative power.

Through either pathway, transgender individuals may face a higher rent burden due to their gender identity. If this increased rental burden leads to increased eviction rates for transgender people, such information may serve to drive further statistical discrimination. As noted earlier, because such statistical discrimination can lead to a systematically lowered bargaining power for transgender individuals in housing markets, a vicious cycle can occur, driving increased rent burdens for transgender individuals and potentially increasing the rate of homelessness among the transgender population.

Data

This paper will use data collected from the Household Pulse Survey (HPS). With the COVID-19 pandemic, the USA began collecting the HPS, which began collecting information about sexual orientation and gender identity (SOGI) of respondents in Summer 2021 and continued until January

⁴ While white individuals may access markdowns for particular home sales, they may also pay net mark-ups to live in neighborhoods which are racially homogeneous and populated by whites. These net mark ups are associated with higher housing prices in white-majority neighborhoods, which tend to be higher income and feature higher quality educational opportunities (Whaley 2024).

2025.⁵ The HPS is a nationally representative survey and the SOGI information is collected without regard to state boundaries, leading to its use by economists interested in the economic status of transgender people (Carpenter et al. 2022).

Respondents to the HPS provide information on behalf of their entire household, which introduces serious challenges to the analysis to follow. Most significantly, respondents are responsible for providing a limited range of demographic information about their household; they identify themselves but provide scant information about other residents of the household (if there are any). Respondents do not provide demographic information about the members of the household, only providing their own demographic information.⁶ This provides a significant barrier to identifying individuals by their identity: racial, gender, or otherwise. Important economic indicators are often aggregated at the household level, too. For example, respondents are asked to provide total household income instead of personal income. Information pertaining to other indicators of economic health, such as mortgage timeliness or utilization of welfare programs, is also collected at the household level.

Despite these challenges, the HPS is still a valuable tool in analyzing the economic status of the USA, especially when considering the economic wellbeing of the populations within it. Aggregated household incomes can often provide a satisfactory understanding of the economic health of members of the household if members of the household share resources. In single-member households, too, aggregation of resources provides information identical to that information, disaggregated. To account for the loss of clarity about the economic status of individuals with aggregation of economic information, the analysis to follow will include estimations which consider only small households, which contain two or one adult.

The following results will use data collected in the HPS during Waves 3.6 through 3.9. Following Carpenter et al. (2022), estimations will employ person-level weights.

⁵ The HPS was replaced by the Household Trends and Outlook Pulse Survey (HTOPS) in January 2025. HTOPS will rely upon a panel design (US Census Bureau 2025). At the time of writing this paper, it is unknown if the HTOPS survey will collect sexual orientation and gender identity information.

⁶ Respondents are asked to identify the number of other individuals in the household, and are asked to divide those other household members by age—those under 18 years old are classified as children, while those above that age are considered adults (respondents are also asked to indicate, given there are children in the household, the presence of children within certain age bins). The HPS also collects information on the military service status of the respondent or other members of the household.

The HPS uses a two-step gender verification process, which is argued to produce more accurate estimations of gender distribution in a population. After supplying their sex assigned at birth—which may be either male or female—survey respondents are asked to provide their gender, where they may identify as either male, female, transgender, or none of these. If an individual replies with a gender which is different from that the respondent's birth sex, they are asked to either confirm this difference or correct their responses.

For this analysis, I use six gender categories which arise from this gender verification process. Cisgender men and women are men and women who were assigned male and female genders at birth and whose gender continues to be male and female, respectively. Transgender women are individuals who were assigned male at birth but who currently identify as a woman or a transgender person. Transgender men are individuals assigned female at birth who identify currently as men or as transgender. Male “nones” are individuals who were assigned male at birth but who do not identify as men, women, or transgender. Similarly, female “nones” were assigned female at birth but do not identify as men, women, or transgender.

These six gender categories differ from those constructed within the HPS, which categorizes respondents as men, women, transgender, or none of these. Even with the two-step gender verification process described above, 4463 respondents responded as identifying as either a woman but assigned a male gender at birth or a man assigned a female gender at birth but not identifying as transgender. Although small compared to the 1,800,997 total unweighted responses, this subset of responses could, if coded as transgender, comprise nearly 40% of the transgender population and could influence the results if they differ systematically from the respondents who answered affirmatively about their transgender status.

I will also exclude individuals who would be identified as transgender due to sex-at-birth imputation. In the HPS, 16,646 respondents had their sex at birth imputed. Individuals with their birth sex imputed were those who had not provided their birth sex when answering the survey, but who, in the construction of the survey weights, had a birth sex assigned to their observation by the Census Bureau (US Census Bureau 2022).

These individuals with birth sex imputed were more likely to be Latine and report lower levels of educational attainment and a higher median age. The income distribution for these individuals was, compared to their peers without imputed sex at birth, far more extreme, with higher proportions reporting household incomes below \$25,000 per year and, for transgender men and male and female “nones,” higher proportions reporting household incomes above \$200,000 per

year. Roughly half of those 4463 individuals who identified as either a woman but assigned a male sex at birth or a man assigned a female sex at birth but not as transgender were individuals whose sex assigned at birth was imputed by the Census Bureau.

As an analysis of race, it is important to clarify the approach I take with regard to racial categories used compared to their deployment by the survey. I employ five racial categories, which are mutually exclusive: white, Black, Asian, Latine, and Any Other Race. An individual in the HPS reports their race and Hispanic status separately, and an individual reporting Hispanic status is classified as Latine, regardless of their identified race. Individuals who do not report a Hispanic ethnicity are classified by the race they identify as, except those who were classified as having “Any Other Race.” These individuals, who may identify as American Indian or Alaska Native, Native Hawaiian, Chamorro, Samoan, or Other Pacific Islander, are included in the subsequent analysis but care must be exercised in analysis using this category: differences among these populations make it logically tenuous to claim the existence of any sort of unified racial effects which are observed in this subset of respondents. Greater data clarity, which can directly identify racial identities contained within this large category, is necessary before coherent claims can be made about the populations of which these respondents are members.

Table 1 shows the demographic characteristics of the sample used. While the most weighed responses come from cisgender men and women, male and female “nones” easily outnumber the transgender population, with 1254 total “nones” compared to 801 total transgender men and women. The sample differs in its demographic composition and these differences tend to divide cisgender men and women from transgender men and women. Meanwhile, male and female “nones” tend to switch, depending on the category, between being similar demographically to cisgender individuals or transgender individuals.

The demographic characteristics of cisgender people are most distinct from those of the rest of the population when considering age, disability status, household rental payments, and achievement of advanced degrees. Cisgender men and women are older than transgender men and women as well as male and female “nones.” The median ages for transgender men and women, as well as for female and male “nones,” are similar to each other. Furthermore, cisgender men and women report far lower rates of disability than other genders. The cisgender population is more likely to earn graduate degrees, compared to other respondents.

The demographic distinction, wherein cisgender people differ from both transgender people and male and female “none,” is not absolute. When considering race, the demographic characteristics of cisgender people are more like those of the female and male “none,” however. Transgender

men and women are far more likely to report being white, compared to other respondent genders.

When considering household economic characteristics, the relative clarity of the earlier patterns vanishes. Cisgender men and male “none” report the highest median income, while all others report an income in the same, lower bin. However, the information provided by this is limited by the use of bins. Cisgender people report higher average household rental payments, compared to individuals of other genders.

Housing Affordability

In the HPS, the amount a household spent on rent was asked in Phases 3.5 to 3.8. While rental payments are listed directly in the HPS, income is reported residing within a range with an upper and lower bound. Using 1, the lower bound for rent burden is calculated using the upper bound of income, $income_{upper}$:

$$rent\ burden_{lower} = \frac{rent}{income_{upper}} \quad (2)$$

while the upper bound for rent burden is calculated using the lower bound of income, $income_{lower}$:

$$rent\ burden_{upper} = \frac{rent}{income_{lower}} \quad (3)$$

By definition:

$$rent\ burden_{lower} < rent\ burden < rent\ burden_{upper} \quad (4)$$

Substituting 4 into 1:

$$\frac{rent}{income_{upper}} < \frac{rent}{income} < \frac{rent}{income_{lower}} \quad (5)$$

With an interval regression, the rent burden for households can be estimated using both the upper and lower bounds of the rent burden. This regression for individual i , responding for their household in state s and during week t , takes the form:

$$rent\ burden_{its} = \beta_0 + Gender_{its}\beta_g + Race_{its}\beta_r + X_{its}\beta_3 + \alpha_s + \gamma_t + \varepsilon_{its} \quad (6)$$

where β_g and β_r are vectors of coefficients modeling the impact of an individual's gender and race, respectively, on rent burden, relative to the base case: a cisgender man,

Table 1 Respondent demographics

	Cisgender man	Cisgender woman	Transgender woman	Transgender man	Male "none"	Female "none"	Total
Median age	35	36	28	27	29	29	35
disabled	12.9%	18.5%	21.0%	32.2%	27.5%	36.2%	16.3%
White	55.5%	53.9%	70.5%	76.2%	52.6%	56.7%	55.0%
Black	11.7%	16.5%	11.2%	3.7%	7.2%	12.5%	13.9%
Asian	6.1%	4.8%	3.1%	1.9%	5.0%	3.2%	5.4%
Latine	22.5%	19.3%	9.3%	12.2%	29.2%	17.1%	20.8%
Any Other Race	4.2%	5.5%	5.8%	6.1%	5.9%	10.5%	5.0%
Some HS or less	6.6%	5.9%	7.7%	0.4%	13.4%	5.0%	6.2%
High school	31.5%	24.1%	30.3%	18.3%	24.4%	22.4%	27.6%
Some college or associates	30.2%	33.6%	39.2%	40.9%	37.5%	39.2%	32.2%
Bachelor's degree	19.6%	21.7%	14.0%	31.3%	15.6%	24.0%	20.8%
Grad degree	12.1%	14.7%	8.8%	9.1%	9.2%	9.4%	13.3%
Median income	\$50,000 - \$74,999	\$35,000 - \$49,999	\$35,000 - \$49,999	\$35,000 - \$49,999	\$50,000 - \$74,999	\$35,000 - \$49,999	\$50,000 - \$74,999
Mean rent	1,547.55	1,482.09	1,387.26	1,419.79	1,358.73	1,449.73	1,510.98
N	28,573	39,885	264	537	375	879	70,513
Weighted N per week	17,433,944	18,049,397	160,044	289,869	277,252	450,199	36,660,705

Table calculated using observations in Phases 3.6–3.9 of the Household Pulse Survey and using person-level weights. A person is classified as disabled if they report great difficulty or an inability to see, hear, remember or concentrate, undertake self-care, understand or be understood by others, or walk or climb stairs. All elements of the sample report renting their housing and provide a non-empty answer to their monthly rental housing payment

with regards to the impact of gender, and a white person, with regards to race. $Gender_{its}$ and $Race_{its}$ are matrices of dummy variables which indicate the gender and race of respondent i in state s and during week t , respectively. Because β_g is a vector of coefficients and $Gender_{its}$ is a vector of dummy variables, their product is the difference in rent burden attributable to the gender of the respondent. The same process is true for differences attributable to respondent racial identity. If there is a positive element in β_g , the average household with a respondent of that gender will be said to pay a rent premium; they pay a higher proportion of household income towards rent, relative to a household with a cisgender man responding to the HPS. The matrix of controls, X_{its} , includes age, age squared, education, marital status, labor force status, disability status, housing type, number of kids, and sector of employment. Week fixed effects γ_t are included to account for wage and rental housing prices changing over time. State fixed effects α_s are included to account for geographic variation in prevailing wage and rental payment distributions, as well as variation in legal frameworks. Standard errors are clustered at the state level.

Rent burden is a relationship between rental payments and income, and changes in rent burden found in subsequent estimations may be the result of differences in either rental payment or income due to transgender identity. Higher (lower) rent burden facing a population may be driven by higher (lower) yearly rent payments or by lower (higher) income. The coincidence of higher (lower) rent payments with higher (lower) yearly income would have an ambiguous influence on rent burden. To decompose the influence of transgender identity onto rent burden into both income and rent payment effects, both will be estimated using separate regressions. Because income is presented as an interval, differences associated with respondent identity are estimated with an interval regression. The interval regression for the household income of individual i in state s and week t takes the form:

$$income_{its} = \beta_{0y} + Gender_{its}\beta_{gy} + Race_{its}\beta_{ry} + X_{its}\beta_{3y} + \alpha_{sy} + \gamma_{ty} + \varepsilon_{itsy} \quad (7)$$

β_{gy} and β_{ry} are vectors of coefficients modeling the influence of respondent gender and race, relative to respondents who are cisgender men and white people, respectively, on household income. Like in Eq. 6, $Gender_{its}$ and $Race_{its}$ are matrices of indicators which classify the race and gender of the respondent. The product of these matrices with vectors β_{gy} and β_{ry} yields average differences in household income attributable to the gender and race of the respondent, relative to cisgender men and white people. The matrix of controls X_{its} contains the same controls as those in Eq. 6. Week and state fixed effects γ_{ty} and α_{sy} are included. Standard errors are clustered at the state level. The use of the subscript y is

to differentiate the elements of the income model from those used in the rent burden model.

Unlike the estimations for rent burden and income, the household rental housing payment is presented as a point, instead of an interval. As such, estimating differences associated with the identity of the respondent does not require an interval regression. The regression for the rent payment for individual i in state s and week t takes the form:

$$Rent\ Payment_{its} = \beta_{0rp} + Gender_{its}\beta_{grp} + Race_{its}\beta_{rrp} + X_{its}\beta_{3rp} + \alpha_{srp} + \gamma_{trp} + \varepsilon_{itsrp} \quad (8)$$

As in Eq. 6 and 7, β_{grp} and β_{rrp} are vectors of coefficients modeling the influence of respondent gender and race on differences in household rental housing payments. The product of matrices $Gender_{its}$ and $Race_{its}$ with β_{grp} and β_{rrp} yields average differences in household rental housing payments attributable to the gender and race of the respondent, relative to cisgender men and white people, respectively. The matrix of controls X_{its} remains the same. Week and state fixed effects γ_{trp} and α_{srp} are included. Standard errors are clustered at the state level. The use of the subscript rp is to differentiate the elements of this model from the income and rent burden model.

Results

Table 2 presents the estimated difference coefficients for gender effects relative to cisgender men on rent burden, income, and rent payments; these are the coefficients composing vectors β_g , β_{gy} , and β_{grp} , respectively. Each estimation controlled for race, education, marital status, employment, age, age squared, disability status, housing type, number of kids, type of work, and each employed state and week fixed effects.

Under the “rent burden” heading are estimated difference coefficients between a given identity category and its reference group. These coefficients were estimated with an interval regression of the form of Equation (6). The second column shows the coefficients of income differences between an identity category and the reference group. This equation takes the form of Equation (7). These coefficients were estimated using an interval regression. The third column shows estimates of differences in yearly rent between an identity group and the reference group, estimated using a linear regression of the form of Equation (8).

First consider the influence of gender, as shown in Table 2. These results find statistically significant evidence that cisgender women, transgender men, and female “nones” live in households which face higher rent burdens than the households of cisgender men. The households of transgender men pay an additional 7.9% of their yearly income towards rental housing, while the households of female “none” pay an

Table 2 Estimations of gender differences in rent burden, income, and yearly rent payments

	Rent burden	Income	Yearly rent
Cisgender woman	0.046*** (0.000)	−13,622.27*** (0.000)	−471.02*** (136.26)
Transgender woman	0.078* (0.043)	−5936.4 (6434.61)	−405.53 (1476.9)
Transgender man	0.079*** (0.018)	−16,093.40*** (4112.97)	−478.24 (461.52)
Male “none”	−0.016 (0.027)	10,881.97 (11,116.39)	−141.23 (744.09)
Female “none”	0.095*** (0.013)	−18,578.72*** (2972.5)	−193.81 (444.29)

*** $p < .01$, ** $p < .05$, * $p < .1$

Coefficients in each row correspond with the difference for the outcome of interest, given by the column header, between the identity category and that of the reference group. Standard errors are in parentheses. The reference group are cisgender men. Further, each regression clusters standard errors at the state level excludes individuals who are recorded as transgender due to imputation of gender at birth, and considers only observations during weeks 49 through 57. Each estimation controls for education, marital status, employment, age, age squared, disability status, housing type, number of kids, type of work, and features state and week fixed effects

additional 9.5%. The estimated difference in rent burden between the households of transgender women and those of the households of cisgender men is positive and significant at a $0.05 < p < 0.10$ level, implying some evidence that transgender women face higher rent burdens than cisgender men. Male “nones” report rent burdens which do not differ from those of cisgender men in any meaningful way, statistical or otherwise.

Although the differences in rent burden between cisgender men and cisgender women, transgender men, and female “nones” are statistically significant, there is insufficient evidence to determine, from differences in rent burden alone, why this estimate was obtained. Estimates of income and household rental payments, which are components of rent burden, can clarify why these differences in rent burden are observed.

Strong and statistically significant evidence suggests that cisgender women live in households reporting lower income (\$17,702 lower, on average) and lower rent payments (\$471 lower) than those of cisgender men. These results, along with earlier evidence suggesting that the households of cisgender women report higher rent burdens than those of cisgender men, imply that the depressed relative incomes found in the households of cisgender women outweigh, in effect, the lower yearly rental housing payments.

For the households of transgender men and female “nones,” the source of the increased rent burden is far more clear: lower average incomes, at the household level, are responsible. The households of transgender men report incomes estimated to be more than \$16,093, on average, lower than households of comparable cisgender men. There is no evidence of differences in average rental housing

payments between the households of cisgender men and transgender men.

Similarly, the households of female “none” are estimated to have incomes which are, on average, \$18,578 lower than that of cisgender men. As with the households of transgender men, there is no evidence of systematic differences in rental housing payments between the households of female “nones” and those of cisgender men.

Neither the households of transgender women nor those of male “nones” show any systematic differences in income or yearly rental payments compared to the households of cisgender men.

The Role of Spatial Heterogeneity

While the HPS does not contain comprehensive information on the location of respondents, it does indicate if respondents live in any of the 15 most populated metropolitan statistical areas (MSAs). These MSAs are listed in Table 3. Because there are hundreds of total MSAs, the HPS can be used to identify a subset of all respondents who may live in an urban area.

Spatial heterogeneity within rental housing markets may influence the results discussed above, particularly if individuals of different genders tend to live in different geographic areas. Research has shown a strong association between gender and sexual diversity and urban areas, though this may also be due to the greater frequency of studies examining gender and sexual diversity in cities (Stone 2018). If urban areas tend to have more expensive housing, the relative clustering of transgender individuals in these cities could contribute to differences in average housing costs.

Table 3 Identifiable MSAs in the Household Pulse Survey

MSA number	Name
12060	Atlanta-Sandy Springs-Alpharetta, GA Metro Area
14460	Boston-Cambridge-Newton, MA-NH Metro Area
16980	Chicago-Naperville-Elgin, IL-IN-WI Metro Area
19100	Dallas-Fort Worth-Arlington, TX Metro Area
19820	Detroit-Warren-Dearborn, MI Metro Area
26420	Houston-The Woodlands-Sugar Land, TX Metro Area
31080	Los Angeles-Long Beach-Anaheim, CA Metro Area
33100	Miami-Fort Lauderdale-Pompano Beach, FL Metro Area
35620	New York-Newark-Jersey City, NY-NJ-PA Metro Area
37980	Philadelphia-Camden-Wilmington, PA-NJ-DE-MD Metro Area
38060	Phoenix-Mesa-Chandler, AZ Metro Area
40140	Riverside-San Bernardino-Ontario, CA Metro Area
41860	San Francisco-Oakland-Berkeley, CA Metro Area
42660	Seattle-Tacoma-Bellevue, WA Metro Area
47900	Washington-Arlington-Alexandria, DC-VA-MD-WV Metro Area

To assess whether spatial heterogeneity is driving observed differences in rent burdens across households with respondents of varying gender identities, I analyze the data using different spatial definitions. This approach will help determine the relevance of location in explaining the differences in rent burden. Specifically, estimated differences observed in the “overall sample”—which includes respondents regardless of location—can be compared to those observed in two sub-samples: the “MSA sample” (households located in the 15 most populous metropolitan statistical areas) and the “non-MSA sample” (households located outside these 15 MSAs). If differences in rent burdens are consistent across both the overall and MSA samples, spatial heterogeneity is less likely to be a contributing factor. In contrast, if differences are only observed in the MSA or non-MSA samples, it suggests that spatial factors may be influencing the results.

As shown in Table 4, only 32.5% of respondents’ households are located in the 15 most populous MSAs. The distribution of respondents varies by gender identity: transgender women and male “nones” were overrepresented in the MSAs compared to the overall sample. Meanwhile, transgender men are slightly underrepresented in the MSAs.

Table 5 presents the estimated differences in the rent burdens, incomes, and rental payments between the households of cisgender men and the households of respondents of other genders. While Table 2 includes all respondents, Table 5 focuses specifically on respondents living in the 15 largest MSAs in the USA. This narrower focus allows Table 5 to show the estimated average differences in rent burden,

Table 4 MSA residency by gender of respondent

	Proportion in the MSA sample
Cisgender men	32.9%
Cisgender women	32.0%
Transgender women	41.5%
Transgender men	31.7%
Male “nones”	37.8%
Female “nones”	33.7%
Overall	32.5%

These proportions were calculated using the HPS-provided person-level weights, and correspond to the proportion of all respondents of a particular gender who indicated their household is located within one of the 15 most populous MSAs in the USA. The Overall column considers all respondents, regardless of gender

income, and yearly rent among residents within these major urban regions.

Within the 15 largest metropolitan statistical areas (MSAs) in the USA, households of cisgender women, transgender men, and female “nones” pay a higher proportion of their income towards rental housing costs compared to households of cisgender men. These estimated differences are consistent with those observed in the broader sample, as shown in Table 2. The households of cisgender women in the MSA sample pay an additional 5.1% of their annual income towards rental housing, compared to 4.6% in the

Table 5 Estimations of gender differences in rent burden, income, and yearly rent payments: MSA only, state, and MSA fixed effects

	Rent burden	Income	Yearly rent
Cisgender woman	0.051*** (0.000)	−14,516.14*** (0.000)	−716.43*** (171.62)
Transgender woman	0.125* (0.066)	−6427.78 (15,377.67)	547.3 (1711.35)
Transgender man	0.176*** (0.041)	−35,822.90*** (6158.42)	−2099.09* (1008.09)
Male “none”	−0.113* (0.062)	−1949.29 (5177.27)	−2835.79* (1600.81)
Female “none”	0.093*** (0.013)	−24,144.85*** (4258.5)	−1080.87 (896.44)

*** $p < .01$, ** $p < .05$, * $p < .1$

Coefficients in each row correspond with the difference for the outcome of interest, given by the column header, between the identity category and that of the reference group. Standard errors are in parentheses. The reference group are cisgender men. Further, each regression clusters standard errors at the state level, excludes individuals who are recorded as transgender due to imputation of gender at birth, and considers only observations during weeks 49 through 57. Each estimation controls for education, marital status, employment, age, age squared, disability status, housing type, number of kids, type of work, and features state, MSA, and week fixed effects

overall sample. Similarly, the difference for female “nones” in the MSA sample remains almost unchanged, with the rent premia increasing from 9.5% in the overall sample to 9.3% in these MSAs.

Like cisgender women and female “nones,” the households of transgender men in the MSA sample also allocate a larger share of their household income towards rental housing payments. However, the difference for transgender men is more pronounced. In the MSA sample, the households of transgender men allocate 17.6% more of household income to rental housing costs, compared to 7.9% in the wider sample.

As with the wider sample, the rent burden disparities for households of cisgender women, transgender men, and female “nones” in the MSA sample are driven by lower average household incomes. In the wider sample, the average household income of cisgender women was \$13,622 less than those of cisgender men. In these MSAs, the difference grows to \$14,516. This income effect partially is offset by lower yearly rent, with the households of cisgender women in MSAs paying \$716 than the households of cisgender men.

The income gap between households of cisgender men and households of both transgender men and female “nones” in the MSA sample is substantially larger than the gap between cisgender men and cisgender women. Specifically, the households of transgender men report incomes which are \$35,822 lower than those of cisgender men—nearly double the income difference observed in the overall sample. There is weak evidence that transgender men in MSAs pay lower rent than cisgender men, though this result is statistically significant at the $0.05 < p < 0.1$ level.

Similarly, the households of female “nones” in the MSA sample also have significantly lower incomes than cisgender men, with an estimated average income difference of \$24,144—roughly a 30% increase compared to the income gap of \$18,578 observed in the overall sample. However, there is no evidence of a difference in rental payments between the households of cisgender men and female “nones.”

For transgender women, weak evidence emerged in both the overall sample and the MSA sample suggesting that their households pay gendered rent premia, though this finding is statistically significant at the $0.05 < p < 0.1$ level.

Finally, weak evidence was found that the households of male “nones” in the MSA sample face lower rent burdens than the households of cisgender men. This is further supported by weak evidence suggesting that the lower rent burden for male “nones” is due to lower annual rent payments. Both findings are statistically significant at the $0.05 < p < 0.1$ level.

Table 6 contains the estimates obtained using the non-MSA sample. As with prior estimates, the households of cisgender women, transgender men, and female “nones”

Table 6 Estimations of gender differences in rent burden, income, and yearly rent payments: only non-MSA, state fixed effects

	Rent burden	Income	Yearly rent
Cisgender woman	0.045*** (0.005)	−12,922.94*** (0.00)	−216.87 (167.41)
Transgender woman	0.022 (0.028)	−6635.27 (4874.47)	−2058.16 (1339.55)
Transgender man	0.045** (0.020)	−7777.55 (5263.55)	361.25 (625.02)
Male “none”	0.034 (0.022)	15,725.8 (15,131.05)	784.67 (641.81)
Female “none”	0.095*** (0.021)	−15,949.22*** (4068.76)	326.17 (464.05)

*** $p < .01$, ** $p < .05$, * $p < .1$

Coefficients in each row correspond with the difference for the outcome of interest, given by the column header, between the identity category and that of the reference group. Standard errors are in parentheses. The reference group are cisgender men. Further, each regression clusters standard errors at the state level, excludes individuals who are recorded as transgender due to imputation of gender at birth, and considers only observations during weeks 49 through 57. Each estimation controls for education, marital status, employment, age, age squared, disability status, housing type, number of kids, type of work, and features state and week fixed effects

exhibit lower average rent burdens than those of cisgender men. The magnitude of the rent burden differences for cisgender women and female “nones” is consistent across both the MSA and overall samples. However, the estimated rent burden difference for transgender men is smaller in the non-MSA sample than in the overall sample, and much lower than in the MSA sample. Additionally, there is no statistically significant difference in rent burden between the households of cisgender men and either transgender women or male “nones.” Weak evidence suggests that the households of transgender women face gendered rent premia, and this finding is observed in both the overall and MSA samples. For male “nones,” although there is no evidence of a rent burden difference in the overall sample, weak evidence indicates that their households may pay a smaller proportion of income toward rent in the MSA sample.

For cisgender women and female “nones,” the decomposition of these results largely confirms earlier findings: the differences in rent burden are driven by lower household incomes. For transgender men’s households, the decomposition does not clearly identify whether the higher rent burden is due to income differences or higher rental payments. As in the overall sample, there is insufficient evidence to suggest that there are systematic differences in income or rental payments between cisgender men and either transgender women or male “nones.”

When considering the estimates across the overall, MSA, and non-MSA samples, it becomes apparent that spatial het-

erogeneity plays a role in the observed differences in rent burden. This heterogeneity is most pronounced for transgender men and women. The rent burden differences between cisgender men and both transgender men and women are larger in magnitude in the MSA sample, followed by the overall sample, and then the non-MSA sample. The most striking example of spatial heterogeneity is the variation in household income differences between cisgender men and transgender men. In the MSA sample, transgender men's households earn \$35,822 less than those of cisgender men, while in the overall sample, the income gap decreases significantly to \$16,093. In the non-MSA sample, this income difference is no longer statistically significant.

In contrast, there is far less evidence of spatial heterogeneity influencing the rent burden differences for cisgender women and female "nones." While there is some variation in household income differences across the samples, the results for these groups are relatively consistent across all spatial definitions.

The Role of Household Structure

Prior analyses have been constrained by the Household Pulse Survey's aggregation of economic data at the household level, which obscures individual-level gender effects on rent burden. To address this limitation, I focus on two specific household types: single-adult households (where household data directly reflects individual circumstances) and two-adult households (where household resource sharing is most likely to occur). This methodological approach provides stronger evidence that transgender individuals themselves face an increased rent burden. Moreover, the results presented below suggest that the HPS's household-level data aggregation may systematically underestimate economic disparities across gender identities, as individual-level financial challenges can be masked when resources are pooled within households.

Table 7 presents the estimated differences in rent burden between households containing one or two adults. Although earlier analyses found evidence suggesting the households of cisgender men may face lower average rent burdens than the households of cisgender women, transgender men, and female "nones," there was mixed evidence of average differences in rent burden between cisgender men, transgender women, and male "nones." When considering only households with two or fewer adults, this mixed evidence is replaced with straightforward and consistent results: the average households for cisgender men face lower rent burdens than those of all other genders. For cisgender women, transgender men, and female "nones," the magnitude of the difference found between household rent burdens remains consistent with those estimated across the entire sample. Meanwhile, there is statistically significant evidence that the households of transgender women and male "nones" face

Table 7 Estimations of gender differences in rent burden, income, and yearly rent payments: singles and couples

	Rent burden	Income	Yearly rent
Cisgender woman	0.037*** (0.000)	−12,855.59*** (0.000)	−746.48*** (94.32)
Transgender woman	0.056** (0.026)	−8341.56** (4121.27)	−1026.26 (819.49)
Transgender man	0.072*** (0.015)	−23,279.91*** (3255.31)	−1836.40*** (374.36)
Male "none"	0.043** (0.020)	−5702.63 (4481.9)	−994.6 (623.33)
Female "none"	0.081*** (0.016)	−19,085.29*** (2228.48)	−789.74** (340.88)

*** $p < .01$, ** $p < .05$, * $p < .1$

Coefficients in each row correspond with the difference for the outcome of interest, given by the column header, between the identity category and that of the reference group. The sample consists of only respondents who reported being the only adult in the household or living with another adult. Standard errors are in parentheses. The reference group are cisgender men. Further, each regression clusters standard errors at the state level, excludes individuals who are recorded as transgender due to imputation of gender at birth, and considers only observations during weeks 49 through 57. Each estimation controls for race, education, marital status, employment, age, age squared, disability status, housing type, number of kids, type of work, and features state and week fixed effects

higher rent burden than those of cisgender men. For transgender women, only weak evidence (statistically significant at the $0.05 < p < 0.1$ level) suggests their households face increased rent burden relative to those of cisgender men. For male "nones," similarly weak evidence had suggested their households paid a smaller percent of yearly income towards rental housing expenses (among those living in the largest MSAs).

Results from the decomposition analysis indicate that these increased household rent burdens are driven by lower household income for cisgender women, transgender men and women, and female "nones." Estimates suggest this upward pressure on household rent burden is mitigated, for the households of cisgender women, transgender men, and female "nones," by lower yearly rental housing payments. However, these lower annual rental housing payments are much smaller in magnitude than the average differences in income observed. The households of cisgender women may pay, on average, \$746 less towards their rental housing, but also report an income which is \$12,855 lower. Female "nones" households have a similar difference in annual rental housing spending, estimated at \$789, but also have a greater decrease in income, at \$19,085. The largest average income difference is between households of cisgender men and transgender men and was estimated at \$23,379. The households of transgender men do report a much greater difference in

yearly rental housing payments, at \$1836, but still report a substantial increase in household rent burden (an increased rent burden of 7.2% of total income, compared to cisgender men).

The decomposition analysis also finds that the increased rent burden of the households of transgender women is driven by lower household incomes, with no evidence of average differences in rental housing payments. The decomposition analysis for male “nones” is inconclusive, with no statistically significant difference between their households and those of cisgender men in either income or rental housing spending.

The information collected by the HPS cannot indicate the economic relationships between members of a household, and the aggregation of economic data can obscure the financial circumstances of transgender individuals when analyzed alongside other household members. To achieve greater analytical precision, I further restrict the sample to single-adult households, where respondent income equals household income. This approach provides the clearest insight into how transgender identity influences rent burden.

Table 8 presents the estimated gender differences for single-adult households. The results reveal significant rent burden disparities between cisgender men and all other gen-

der identities except male “nones.” Cisgender women face increased average rent burdens of 4.8%, consistent with prior estimates. Transgender men and women both face increased rent burden. Transgender women pay 9.4% more of their annual income to rental housing, while transgender men pay 10.5% more than comparable cisgender men. Female “nones” also experience a rent burden premium, paying 5.7% more per year than similar cisgender men.

Decomposition analysis indicates that, as observed throughout this study, lower incomes among non-cisgender men primarily drive these higher rent burdens. With the exception of transgender women, there is evidence that cisgender men actually pay higher absolute amounts for rental housing on average. However, these higher housing costs do not offset the substantial income disparities across gender identities.

Male “nones” are a notable exception in this analysis. While they show no significant difference in overall rent burden compared to cisgender men, decomposition reveals that male “nones” face both lower income and lower rental housing costs than cisgender men, with these effects effectively canceling each other out.

What About Home Owners?

It is important to note that rent burden represents just one dimension of housing affordability and excludes homeownership households. Appendix B examines gender differences in timely housing payments for homeowner and renter households. Though I find that cisgender women are more likely than cisgender men to report timely rental payments, no other statistically significant gender differences emerge for either rental or mortgage payments timeliness. This finding suggests that households facing increased rent burden may maintain housing payment timeliness by sacrificing other expenditures.

Evidence of Racial Heterogeneity

An emerging subject of study within economics is that of intersectionality, which highlights the heterogeneity in the influence of identity which arises in the interaction of differing types of identities (Banks 2021). Intersectional methods are closely tied with the emerging “economics of transgender individuals” literature. Carpenter et al. (2022) find evidence of intersectional influences of race and gender on labor market outcomes and impoverishment. Further, Campbell and Rodgers (2023) test for but find no evidence of racial heterogeneity with regards to the harm caused by conversion therapy. The importance of intersectional effects is also

Table 8 Estimations of gender differences in rent burden, income, and yearly rent payments: singles

	Rent burden	Income	Yearly rent
Cisgender woman	0.046*** (0.000)	−13,019.33*** (2357.60)	−458.57*** (168.99)
Transgender woman	0.094** (0.046)	−10,252.05** (4657.87)	400.08 (737.89)
Transgender man	0.105*** (0.037)	−17,800.61*** (3446.74)	−1074.32* (624.64)
Male “none”	0.014 (0.025)	−12,471.20*** (3148.94)	−1698.32*** (516.79)
Female “none”	0.057*** (0.015)	−16,431.94*** (3712.56)	−1282.10*** (445.51)

*** $p < .01$, ** $p < .05$, * $p < .1$

Coefficients in each row correspond with the difference for the outcome of interest, given by the column header, between the identity category and that of the reference group. The sample consists of only respondents who reported being the only adult in the household. Standard errors are in parentheses. The reference group are cisgender men. Further, each regression clusters standard errors at the state level, excludes individuals who are recorded as transgender due to imputation of gender at birth, and considers only observations during weeks 49 through 57. Each estimation controls for education, marital status, employment, age, age squared, disability status, housing type, number of kids, type of work, and features state and week fixed effects

recognized, if implicitly, within the housing discrimination literature. Flage (2018) notes ethnic heterogeneity in gender discrimination, as mentioned above; though all women report lessened discrimination relative to men, ethnic minority women see an even larger decrease in discrimination relative to ethnic minority men than ethnic majority women do compared to ethnic majority men.

The intersection of identity may also lead to unexpected results. Robinson (2002) notes that mortgage lenders may lend less often to white families with a woman who works compared to a woman staying home to raise children but also punish Black or Latine women who care for children compared to Black and Latine women who work. This result emerges from the racialization of motherhood, Robinson argues, where the motherhood of white people is encouraged while, for Black and Latine women, that motherhood is discouraged; the race of mothers determines whether a patriarchal role, like child-rearing, is socially beneficial or socially harmful. The confluence of gender and race influence labor market outcomes, as well. Studying tipped incomes, Compton and Compton (2024) find that white women benefit from discriminatory customer behavior. Meanwhile, tipped incomes for racial minorities, and especially for Black men and women, suffer. These results, due to the structure of tipped minimum wage laws (wherein firms must provide income up to the level of the non-tipped minimum wage), firms are encouraged to hire white women and to not hire minoritized men and women. The established complex interplay of race and gender implies, then, the intersection of transgender identities with race may influence the importance or salience of gender identities across racial groups.

Given the possibility of complex results arising from the confluence of racial and gender identity, it is important to determine if there is evidence of racial heterogeneity in the influence of gender on housing affordability. Each of the above estimations of rent burden controlled for racial identity. The results of the estimation using the overall sample (which considers respondents of all household sizes and households in the overall sample, the MSA sample, and the non-MSA sample) are presented in Table 9. The Rent Burden, Income, and Yearly Rent columns present the estimated coefficients composing vectors β_r , β_y , and β_{rrp} , respectively, as well as the standard errors of the estimated coefficients. The estimations of rent burden and its decomposition for MSAs, non-MSAs, and different household sizes are contained in Appendix A.

Across specifications, the results were relatively consistent: there is strong evidence suggesting the households of white respondents faced a lower rent burden than those of respondents who are Black, Asian, Latine, or of Any Other

Table 9 Estimations of racial differences in rent burden, income, and yearly rent payments

	Rent burden	Income	Yearly rent
Black	0.048*** (0.000)	−17,702.23*** (0.000)	−1111.96*** (262.96)
Asian	0.049*** (0.015)	−3300.56 (2139.86)	219.62 (403.06)
Latine	0.052*** (0.010)	−17,453.81*** (2131.59)	−575.05* (295.95)
Any Other Race	0.019*** (0.006)	−7814.10*** (1214.09)	−52.7 (228.55)

*** $p < .01$, ** $p < .05$, * $p < .1$

Coefficients in each row correspond with the difference for the outcome of interest, given by the column header, between the identity category and that of the reference group. Standard errors are in parentheses. The reference group are white people. Further, each regression clusters standard errors at the state level, excludes individuals who are recorded as transgender due to imputation of gender at birth, and considers only observations during weeks 49 through 57. Each estimation controls for education, marital status, employment, age, age squared, disability status, housing type, number of kids, type of work, and features state and week fixed effects

Race.⁷ This result is driven by differences in household income, though households with white respondents often had higher average rental housing costs to mitigate this income effect.⁸

As discussed earlier, evidence suggests that gender influences on rental market outcomes may be heterogeneous with regard to race. Evidence presented below suggests that the influence of gender—and especially transgender status—on rent burden may be heterogeneous. To test for the possibility of racial heterogeneity in the influence of gender on household rent burden, the regressions specified in Eq. 6 can be constrained to specific racial groups. The estimation for the rent burden for a household with respondent i of race j in state s during week t takes the form:

$$\text{rent burden}_{ijts} = \beta_{0j} + \text{Gender}_{ijts}\beta_{gj} + X_{ijts}\beta_{3j} + \alpha_{sj} + \gamma_{tj} + \varepsilon_{ijts} \quad (9)$$

⁷ There is not statistically significant evidence of differences in household rent burden between single-adult households with a white respondent and those with a respondent who is of Any Other Race.

⁸ The exception to this is in single-adult households with an Asian respondent. Though there is evidence that the single-adult households with Asian respondents face higher yearly rent burden compared to similar white respondents, single-adult households with Asian respondents were estimated to earn higher incomes and also pay higher yearly rental housing payments.

Table 10 Estimation of rent burden, conditional on race: sample size

	White	Black	Asian	Latine	Any Other Race
Cisgender man	20,104	1900	2043	3156	1370
Cisgender woman	26,377	4948	1682	4647	2231
Transgender woman	200	10	10	25	19
Transgender man	414	17	9	62	35
Male “none”	260	19	25	43	28
Female “none”	607	69	33	92	78
Total	47,962	6963	3802	8025	3761

Cells give the weighted sample size used when estimating differences in rent burden conditional on the race of the respondent, shown in Table 11. These weighted sample sizes employ the person-level weights provided by the HPS

Equation 9 serves the same function within racial categories as Eq. 6 offers across racial boundaries. The results of interest, the estimated coefficients constituting the matrix β_{gj} , allow for the influence of gender to vary by racial category. Because the sample is limited to households with respondents of a specific race, the coefficient β_{gj} will capture differences in household rent burden between a cisgender man of race j and a respondent with a different gender, but also of race j . Differences in the values of these β_{gj} coefficients across racial groups will indicate heterogeneity between racial groups as to the influence of identity factors on rent burden. Differences in the influence of transgender status across racial categories are of particular interest. The model includes the same suite of control variables and also includes state and week fixed effects. Errors are, as in prior estimations, clustered at the state level.

Table 10 highlights a crucial limitation for the following analysis. When examining gender effects within specific racial categories, the sample sizes decrease substantially for certain gender identities. These reduced sample sizes have important statistical implications: they increase standard errors, which in turn makes it more difficult to achieve statistical significance even when meaningful gender effects may exist. Consequently, this analysis may fail to detect genuine gender-based disparities due to insufficient statistical power rather than their absence in reality.

Table 11 presents the estimated differences in rent burden between a cisgender man and a person with a different gender but contingent on the race of the individual. As such, differences in the influence of gender across racial groups are indicative of the heterogeneous influence of gender across racial groups.

Table 11 Estimations of rent burden, conditional on race

	White	Black	Asian	Latine	Any Other Race
Cisgender woman	0.031*** (0.007)	0.082*** (0.000)	0.006*** (0.000)	0.077*** (0.012)	0.047*** (0.010)
Transgender woman	0.049** (0.024)	0.255 (0.182)	0.052 (0.101)	−0.06 (0.072)	0.038 (0.058)
Transgender man	0.050*** (0.017)	0.012 (0.053)	0.363 (0.258)	0.189*** (0.073)	−0.01 (0.088)
Male “none”	−0.007 (0.025)	0.041 (0.029)	−0.041 (0.051)	−0.062 (0.061)	0.075 (0.081)
Female “none”	0.082*** (0.018)	0.054* (0.032)	0.028 (0.037)	0.153*** (0.054)	0.085 (0.054)

*** $p < .01$, ** $p < .05$, * $p < .1$

Coefficients in a particular row are the estimates of rent burden associated with the difference between an individual of a specific gender and that of the reference group, cisgender men, within a specific racial group, given by the column. Standard errors are in parentheses. The rent burden is estimated using an interval regression which controls for education, marital status, employment, age, age squared, disability status, housing type, number of kids, and type of work. Further, the interval regression includes state and week fixed effects, clusters standard errors at the state level, excludes individuals who are recorded as transgender due to imputation of gender at birth, and considers only observations during weeks 49 through 57

The evidence reveals distinct patterns of gendered rent burden differences across racial groups. Households of cisgender women consistently face higher rent burdens than those of cisgender men within their racial group. When considering transgender respondents, I find evidence that the households of white transgender women, white transgender men, and Latine transgender men face higher rent burdens than cisgender men within the same racial categories. These disparities in rent burdens can be primarily attributed to differences in household income. These differences are presented in Table 17.

The income data supports this conclusion. With the exception of cisgender Asian women, cisgender women across all racial categories report lower household incomes than cisgender men of their same race. Among white respondents, transgender women, transgender men, and female “nones” all report significantly lower household incomes compared to white cisgender men. A similar income disadvantage is observed for Latine female “nones.”

There are elements of the conditional income and rental payment decompositions which indicate there exist rent burden differences within a racial category and across genders, even if there was no estimated rent burden difference in Table 11. Due to limited sample sizes, it would be methodologically unsound to make definitive claims about “true” rent burden coefficients when no statistically significant coefficient was estimated in either the rent burden estimation or in other elements of the decomposition. Such claims would require assumptions about the “true” value of other decomposition components, potentially introducing bias.

Still, these results provide evidence of racial heterogeneity in gender effects. This is most evident when comparing rent burden differentials between cisgender women and men across racial groups. For instance, the estimated difference between Black cisgender women and men substantially exceeds that between white cisgender women and men.

The data consistently shows that rent burden differences attributable to gender identity are considerably larger for Latine individuals compared to white individuals of the same gender. However, I must acknowledge the limitations in drawing comprehensive conclusions from these results due to the relatively large standard errors associated with rent burden differences; more data must be collected to draw further conclusions.

Conclusion

This research, using the Household Pulse Survey, demonstrates that transgender individuals and females who identify

as neither cisgender nor transgender experience a higher housing cost burden compared to cisgender men.

Significant spatial heterogeneity exists in the relationship between gender identity and household rent burden. The disparity is particularly pronounced in the 15 most populous MSAs in the USA, where the households of transgender men face rent premia approximately three times higher (17.6%) than their counterparts outside these metropolitan areas (4.5%). In contrast, households of cisgender women and female “nones” show rent premia consistent across different geographic contexts. The evidence regarding rent premia for households of transgender women and male “nones” is less conclusive, with weak indications that transgender women in populous MSAs face rent burden premia, while male “nones” may experience slight rent burden discounts relative to cisgender men.

A methodological limitation of the Household Pulse Survey is its collection of economic data at the household rather than individual level. To address this limitation, I restricted the sample to households with two or fewer adults, providing clearer insights into respondents’ economic circumstances. In this more focused analysis, respondent gender identity consistently correlates with household rent premia across all gender identities, with particularly strong effects in single-adult households. The reduced magnitude of gender effects in larger households suggests that individuals may increase household size as a strategy to manage lower personal income.

Our analysis of rent premia within racial groups reveals heterogeneous gender effects. Consistent with the broader findings, cisgender women across all racial categories faced rent premia compared to cisgender men of their race. White and Latine transgender men experienced significant rent premia compared to cisgender men of the same racial background. Similarly, White and Latine female “nones” faced rent premia, with weaker evidence suggesting Black female “nones” may experience similar disadvantages.

These differential rent premia across racial groups suggest that gender-based economic disparities manifest differently according to race. Drawing parallels to research on racial discrimination, this pattern may indicate bargaining power imbalances in housing markets, where populations with less leverage pay premium rates that effectively transfer resources to more advantaged groups.

While this data cannot definitively establish whether transgender and gender non-conforming individuals face discrimination in housing markets, the persistent rental housing payment premia observed is consistent with such discrimination.

Appendix A Tables and Figures

Table 12 Estimations of racial differences in rent burden, income, and yearly rent payments: MSA Only, state and MSA fixed effects

	Rent burden	Income	Yearly rent
Black	0.054*** (0.000)	−27,416.78*** (0.000)	−2552.20*** (685.78)
Asian	0.06*** (0.000)	−16,460.54*** (0.000)	−1738.47*** (529.88)
Latine	0.055*** (0.000)	−25,785.66*** (0.000)	−2286.76*** (581.86)
Any Other Race	0.025*** (0.000)	−13,526.63*** (0.000)	−912.81 (569.45)

*** $p < .01$, ** $p < .05$, * $p < .1$

Coefficients in each row correspond with the difference for the outcome of interest, given by the column header, between the identity category and that of the reference group. Standard errors are in parentheses. The reference group are white people. Further, each regression clusters standard errors at the state level, excludes individuals who are recorded as transgender due to imputation of gender at birth, and considers only observations during weeks 49 through 57. Each estimation controls for education, marital status, employment, age, age squared, disability status, housing type, number of kids, type of work, and features state, MSA, and week fixed effects

Table 13 Estimations of racial differences in rent burden, income, and yearly rent payments: only non-MSA, state fixed effects

	Rent burden	Income	Yearly rent
Black	0.037*** (0.008)	−15,501.90*** (2582.77)	−989.70*** (218.11)
Asian	0.024*** (0.009)	−1390.01 (1493.57)	396.75 (478.26)
Latine	0.028*** (0.006)	−14,504.78*** (2637.14)	−500.71* (262.56)
Any Other Race	0.012* (0.007)	−6935.00*** (1012.96)	−61.93 (247.03)

*** $p < .01$, ** $p < .05$, * $p < .1$

Coefficients in each row correspond with the difference for the outcome of interest, given by the column header, between the identity category and that of the reference group. Standard errors are in parentheses. The reference group are white people. Further, each regression clusters standard errors at the state level, excludes individuals who are recorded as transgender due to imputation of gender at birth, and considers only observations during weeks 49 through 57. Each estimation controls for education, marital status, employment, age, age squared, disability status, housing type, number of kids, type of work, and features state and week fixed effects

Table 14 Estimations of racial differences in rent burden, income, and yearly rent payments: singles and couples

	Rent burden	Income	Yearly rent
Black	0.039*** (0.004)	−16,566.88*** (0.000)	−1075.58*** (215.74)
Asian	0.037*** (0.012)	179.12 (1796.23)	141.69 (242.09)
Latine	0.041*** (0.008)	−14,287.24*** (1689.41)	−771.40*** (237.77)
Any Other Race	0.013** (0.005)	−6624.72*** (1124.74)	−394.58 (238.73)

*** $p < .01$, ** $p < .05$, * $p < .1$

Coefficients in each row correspond with the difference for the outcome of interest, given by the column header, between the identity category and that of the reference group. Standard errors are in parentheses. The reference group are white people. Further, each regression clusters standard errors at the state level, excludes individuals who are recorded as transgender due to imputation of gender at birth, and considers only observations during weeks 49 through 57. Each estimation controls for education, marital status, employment, age, age squared, disability status, housing type, number of kids, type of work, and features state and week fixed effects

Table 15 Estimations of racial differences in rent burden, income, and yearly rent payments: singles

	Rent burden	Income	Yearly rent
Black	0.024*** (0.000)	−8327.38*** (1425.02)	−620.31*** (182.04)
Asian	0.022*** (0.000)	5935.90** (2998.50)	1038.87** (404.30)
Latine	0.031*** (0.010)	−7532.03*** (1644.49)	−456.08 (347.63)
Any Other Race	0.013 (0.009)	−4729.52*** (1385.12)	−296.82 (214.46)

*** $p < .01$, ** $p < .05$, * $p < .1$

Coefficients in each row correspond with the difference for the outcome of interest, given by the column header, between the identity category and that of the reference group. Standard errors are in parentheses. The reference group are white people. Further, each regression clusters standard errors at the state level, excludes individuals who are recorded as transgender due to imputation of gender at birth, and considers only observations during weeks 49 through 57. Each estimation controls for education, marital status, employment, age, age squared, disability status, housing type, number of kids, type of work, and features state and week fixed effects

Table 16 Gender income differences by race

	White	Black	Asian	Latine	Any Other Race
Cisgender woman	−13,952.86*** (0.00)	−16,692.08*** (3082.89)	−2536.59 (1797.69)	−14,481.28*** (0.00)	−9722.67** (3784.82)
Transgender woman	−11,348.34** (5211.91)	−7241.27 (11,315.06)	−17,909.22 (20,026.88)	16,835.51 (20,956.89)	−24,426.26** (12,365.30)
Transgender man	−19,619.05*** (4098.34)	−3527.2 (7922.75)	−43,076.43*** (14,078.35)	−8464.41 (17,250.16)	11985.8 (23,174.07)
Male “none”	−3683.63 (5431.75)	3520.05 (10,333.34)	19,488.07 (13,383.25)	31,851.64 (26,626.98)	4732.6 (9187.23)
Female “none”	−21,042.87*** (2289.41)	−14,335.78** (6680.04)	−23,279.05* (13,442.71)	−19,849.93** (8784.31)	−6617.53 (5558.96)

*** $p < .01$, ** $p < .05$, * $p < .1$

Coefficients in a particular row are the estimates of income associated with the difference between an individual of a specific gender and that of the reference group, cisgender men, within a specific racial group, given by the column. Standard errors are in parentheses. The income is estimated using an interval regression which controls for education, marital status, employment, age, age squared, disability status, housing type, number of kids, type of work. Further, the interval regression includes state and week fixed effects, clusters standard errors at the state level, excludes individuals who are recorded as transgender due to imputation of gender at birth, and considers only observations during weeks 49 through 57

Table 17 Gender rent payment differences by race

	White	Black	Asian	Latine	Any Other Race
Cisgender woman	−489.46*** (155.92)	−555.62** (229.83)	401.77 (378.59)	−414.07 (426.21)	−781.12* (393.23)
Transgender woman	−863.3 (1387.3)	5025.29*** (1235.71)	−2274.29 (2782.12)	−3897.36 (3833.25)	−974.55 (2065.32)
Transgender man	−745.73 (532.92)	668.46 (2806.01)	1390.82 (5541.67)	378.41 (1444.64)	−2254.12* (1290.05)
Male “none”	−1266.21 (852.1)	−1618.84 (1127.74)	9112.47*** (2286.84)	288.48 (984.71)	642.03 (1462.92)
Female “none”	−774.65 (701.43)	287.87 (1770.02)	−1579.85 (1359.07)	182.81 (2204.57)	1106.7 (1097.33)

*** $p < .01$, ** $p < .05$, * $p < .1$

Coefficients in a particular row are the estimates of yearly rent associated with the difference between an individual of a specific gender and that of the reference group, cisgender men, within a specific racial group, given by the column. Standard errors are in parentheses. The yearly rent difference is estimated using a linear regression which controls for education, marital status, employment, age, age squared, disability status, housing type, number of kids, and type of work. Further, the regression includes state and week fixed effects, clusters standard errors at the state level, excludes individuals who are recorded as transgender due to imputation of gender at birth, and considers only observations during weeks 49 through 57

Appendix B Payment Delinquency

This measure of rent burden as an indication of the degree to which transgender people are less able to afford housing is limited, however, to considering only renters and this data was only collected for a relatively short period of time. As such, it will be worthwhile to complement analysis of rent burden with a second measure of housing affordability: payment delinquency. A respondent to the HPS who has been unable to pay for their mortgage or rental payments is considered delinquent. While the HPS has not collected payment delinquency data in each of its iterations, it has been con-

sistently collected since Phase 3.2, which coincides with the beginning of SOGI information collection. For consistency with the earlier estimates of rent burden, however, this analysis will apply data collected in Phases 3.6–3.9.

The influence of gender on payment delinquency for the household of the respondent in a given state and week will be estimated using a probit model. Like prior models used in this paper, the variables of interest are differences attributable to gender. As such, I report the estimated average marginal effects of gender identity on the likelihood of a respondent reporting both rental housing payment and mortgage payment delinquency in Table 18. This model includes the same

Table 18 Average marginal effect of respondent gender on rent and mortgage timeliness

	Rent timeliness	Mortgage timeliness
Cisgender woman	0.007*** (0.002)	0.003* (0.002)
Transgender woman	0.018 (0.015)	−0.002 (0.013)
Transgender man	−0.004 (0.007)	0.043 (0.030)
Male “none”	0.001 (0.008)	0.035 (0.037)
Female “none”	0.012 (0.009)	−0.002 (0.006)

*** $p < .01$, ** $p < .05$, * $p < .1$

Coefficients in each row correspond with the average marginal effect associated with the gender of respondent on the probability the household reports timely payment of either rent or mortgage, given by the column header. Standard errors are in parentheses. The reference group are cisgender men. Further, each regression clusters standard errors at the state level, excludes individuals who are recorded as transgender due to imputation of gender at birth, and considers only observations during weeks 49 through 57. Both control for race, education, marital status, employment, age, age squared, disability status, housing type, number of kids, type of work, and features state and week fixed effects

controls used throughout this paper: age, age squared, education, marital status, labor force status, disability status, housing type, number of kids, and sector of employment. The model also includes controls for the racial identity of the respondent, the state where the respondent lives, and the week of the survey.

The results presented in Table 18 show estimated average marginal effects associated with gender on household rental payment and mortgage payment timeliness. As shown, there is statistically significant evidence that the households of cisgender women are more likely than those of cisgender men to report timely rental housing payment. There is insufficient evidence to suggest the households of cisgender men differ meaningfully in their rental housing or mortgage payments, relative to those of transgender women, transgender men, male “nones,” or female “nones.”

Author Contributions All work done by Rob Haggard

Data Availability All data is publicly available from the US Census Bureau: <https://www.census.gov/programs-surveys/household-pulse-survey/data/datasets.html>

This research uses Phases 3.6–3.9 of the Household Pulse Survey.

Declarations

Ethical Approval and Consent to Participate Not applicable

Human Ethics Not applicable

Consent for Publication Not applicable

Competing Interests The author declares no competing interests.

Clinical Trial Number Not applicable

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